Bike Sharing Demand Forecasting: OLS, Ridge, and Lasso Analysis

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Abstract

This report analyses daily bike rental demand using the UCI Bike Sharing Dataset, comprising 731 observations collected from the Capital Bikeshare system in Washington, D.C. between 2011 and 2012. We apply Ordinary Least Squares (OLS) regression as the primary model, and compare it against Ridge and Lasso regression to evaluate the effect of regularisation. Following careful feature selection and multicollinearity diagnostics, the final model is trained on 8 predictors spanning temporal, weather, and calendar dimensions.

The OLS model achieves an in-sample R^2 of 0.827 and a 5-fold cross-validated RMSE of 821, outperforming Ridge (835) and Lasso (831). Year and temperature emerge as the strongest predictors. Residual diagnostics confirm that linear model assumptions are largely satisfied, with only mild heteroscedasticity observed at higher fitted values.

1 Introduction

Bike-sharing systems have become an increasingly popular form of urban transportation, offering a flexible and environmentally friendly alternative to private vehicles. Accurately forecasting daily rental demand is valuable for operators seeking to optimise fleet distribution, staffing, and infrastructure planning. Demand is inherently influenced by a combination of weather conditions, seasonal patterns, and calendar effects, making it a well-suited problem for regression modelling.

This project analyses daily bike rental counts using the UCI Bike Sharing Dataset, which records 731 days of rental activity across 2011 and 2012 in Washington D.C. The dataset contains weather measurements, temporal indicators, and usage statistics collected by the Capital Bikeshare system. We apply Ordinary Least Squares (OLS) regression as the primary model for its interpretability, and compare it against Ridge and Lasso regression to evaluate whether regularisation offers predictive gains under a well-specified feature set. Through exploratory data analysis, correlation screening, and variance inflation factor (VIF) diagnostics, we carefully remove redundant and aliased predictors before modelling. We further assess the validity of linear model assumptions using residual diagnostics, and report 5-fold cross-validated RMSE as the primary evaluation metric.

2 Data

2.1 Dataset Overview

The dataset contains 731 daily observations collected between January 2011 and December 2012 from the Capital Bikeshare system. It comprises 16 variables, including temporal indicators, weather measurements, and rental usage counts. No missing values were found. The target variable *cnt* represents the total number of bike rentals per day, combining both casual and registered users.

2.2 Feature Description

The dataset contains 16 variables: 2 identifiers (*instant*, *dteday*, excluded from analysis), 1 continuous target (*cnt*), and 13 predictors. Features span temporal (*season*, *yr*, *mnth*, *holiday*, *weekday*, *workingday*), weather (*weathersit*, *temp*, *atemp*, *hum*, *windspeed*), and usage (*casual*, *registered*) dimensions. Among the 13 predictors, 6 are categorical and 5 are continuous.

2.3 Feature Selection and Preprocessing

Prior to modelling, four variables were removed from the dataset. *instant* and *dteday* are excluded as they serve purely as a record index and a date string respectively, carrying no predictive information. *casual* and *registered* are excluded due to target leakage: they are direct components of *cnt* (i.e. $cnt = casual + registered$), and including them would trivially inflate model performance without providing genuine predictive value. Further variables (*atemp*, *mnth*, *workingday*) are examined and removed in subsequent sections based on correlation analysis and multicollinearity diagnostics.

3 Exploratory Data Analysis

3.1 Feature Distribution

Figure 1 shows the distribution of daily rental counts. Counts range from near zero to approximately 8,700. The distribution is roughly symmetric but exhibits a slight secondary peak around 7,000–7,500, likely reflecting the higher overall demand in 2012 compared to 2011.

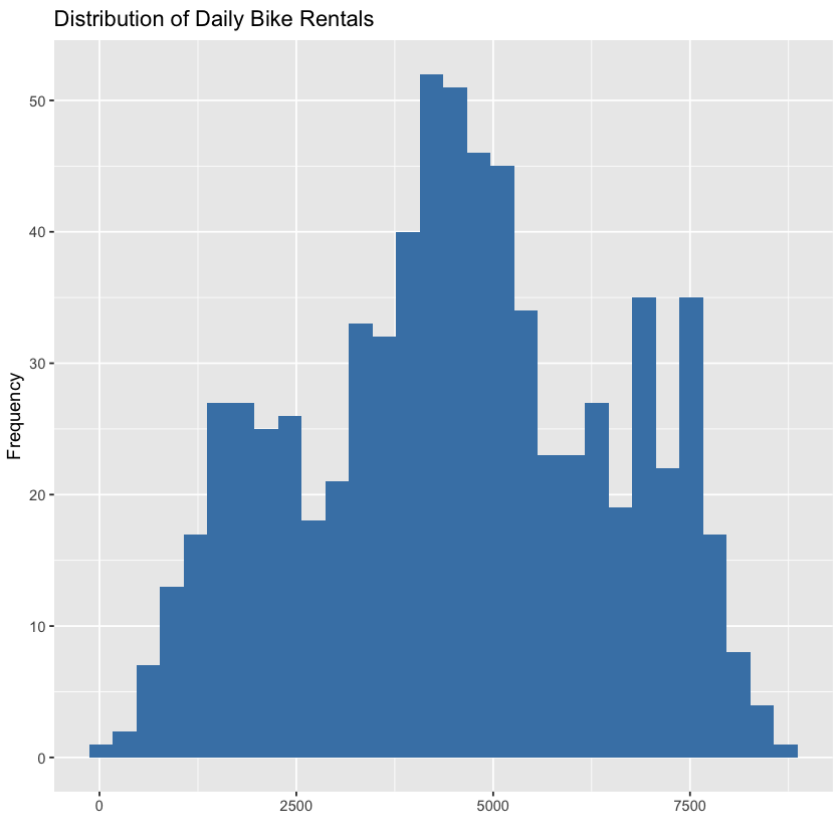

Figure 1: Distribution of Daily Bike Rentals

3.1.1 Categorical Features

Figures 2 and 4 show consistent seasonal trends: counts peak during summer (months 6–9) and drop substantially in winter (months 1–2), with spring and fall occupying intermediate levels. The two plots convey near-identical distributional patterns, reflecting the strong overlap between the two variables. Figure 3 shows a marked increase in overall demand from 2011 (Year 0) to 2012 (Year 1), with the median rising from approximately 3,500 to 6,200. A small number of outliers are visible across all three plots, representing isolated days with unusually low counts; these are consistent with extreme weather events and are considered a normal occurrence in the data.

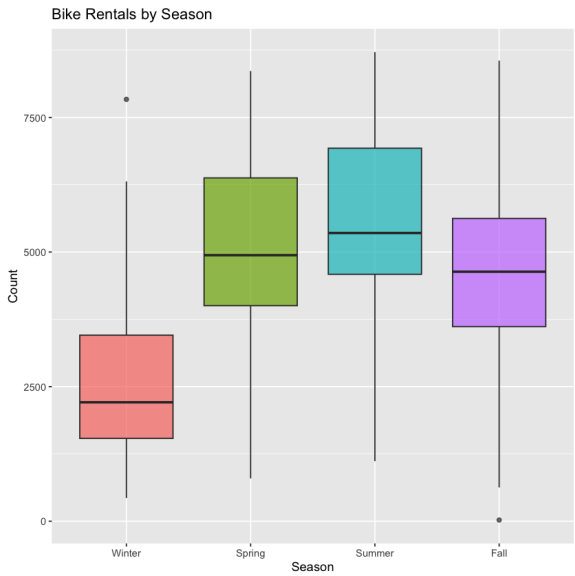

Figure 2: Bike Rentals by Season

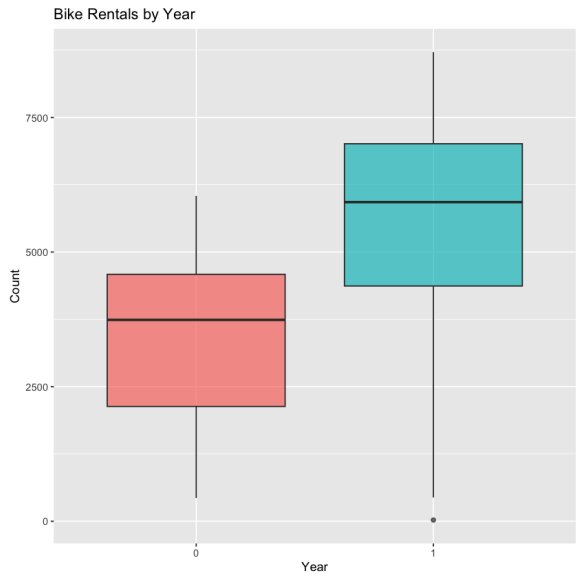

Figure 3: Bike Rentals by Year

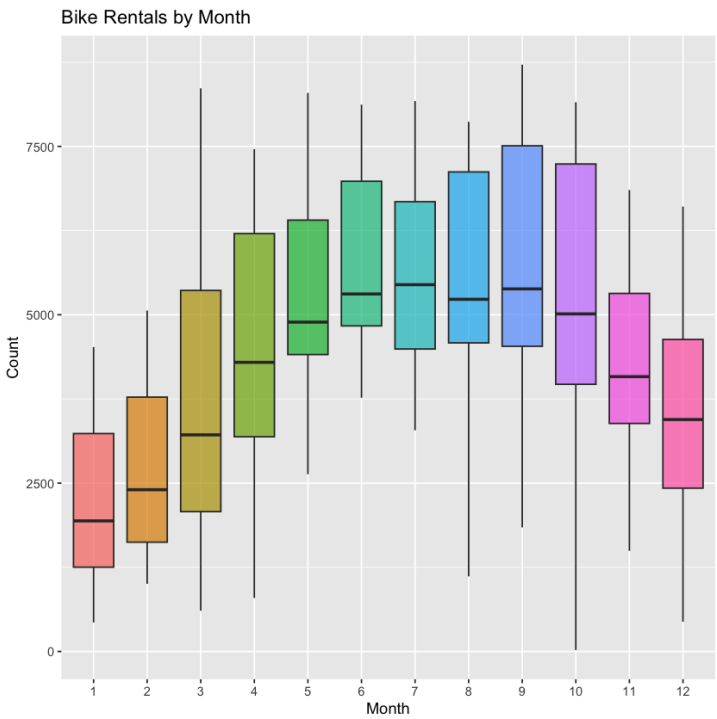

Figure 4: Bike Rentals by Month

Figure 5 shows rental counts across weather conditions and day-type indicators. *weathersit* shows a clear negative trend: higher values indicate more severe weather, and rental counts decline accordingly. Level 4 does not appear in the dataset as no such extreme weather days were recorded. *holiday*, *weekday*, and *workingday* show largely overlapping distributions with no strong separation between categories.

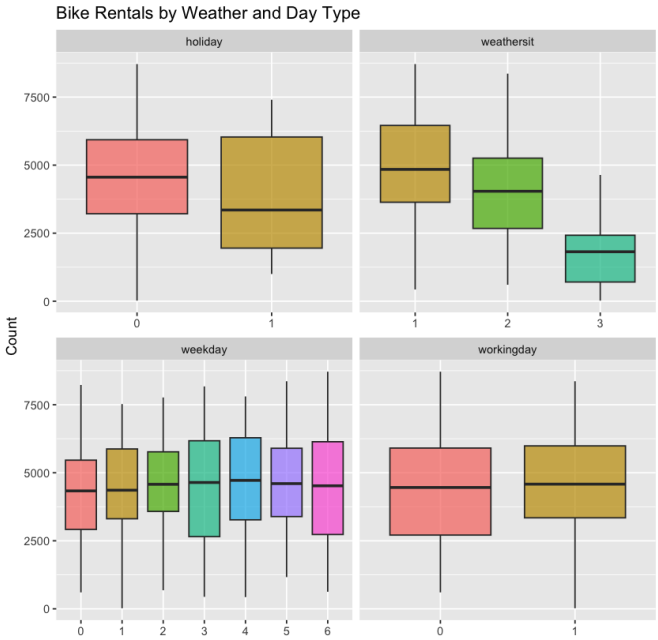

Figure 5: Bike Rentals by Weather and Day Type

3.1.2 Numeric Features

Figure 6 shows the relationship between each numeric feature and daily rental counts. Both *temp* and *atemp* display near-identical positive trends, suggesting that the two variables carry largely redundant information. *hum* and *windspeed* both show negative relationships with rental counts, indicating that higher humidity and stronger winds are associated with fewer rentals.

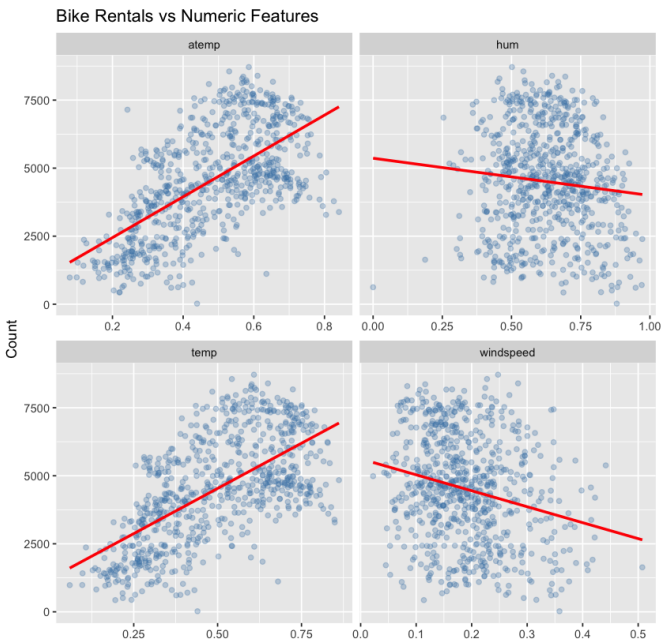

Figure 6: Bike Rentals vs Numeric Features

3.2 Feature Correlation

Figure 7 shows the correlation matrix for numeric features. The most notable finding is the near-perfect correlation between *temp* and *atemp* ($r = 0.99$), confirming the visual observation from the scatter plots and providing a clear justification for removing one of the two variables. Both *hum* ($r = -0.10$) and *windspeed* ($r = -0.23$) show negative correlations with *cnt*, consistent with the trends observed earlier.

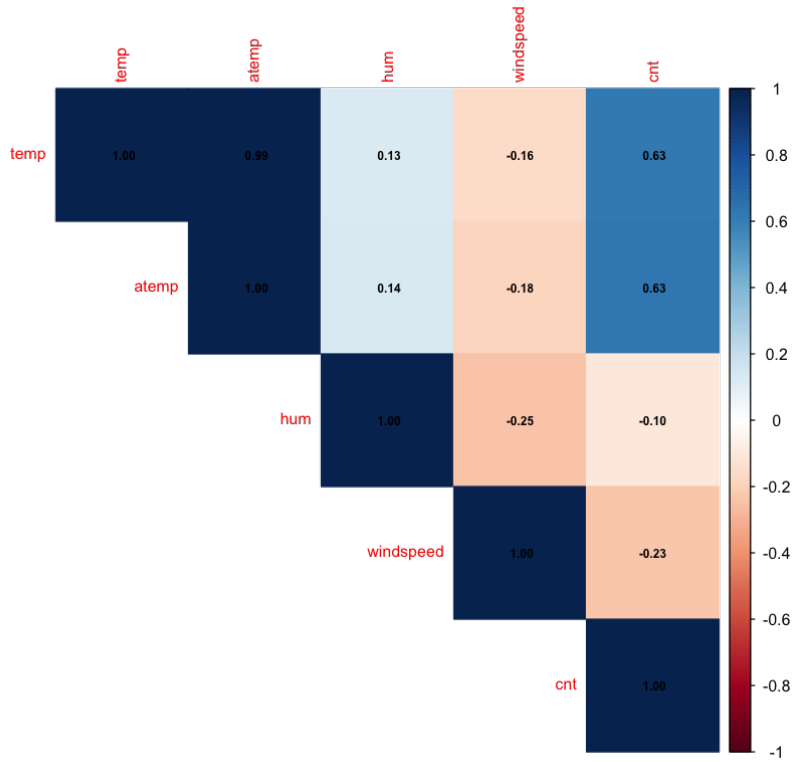

Figure 7: Correlation Matrix of Numeric Features

3.3 Multicollinearity Check (VIF)

Variance Inflation Factor (VIF) was computed to assess multicollinearity among the remaining predictors. Before running VIF, three variables were excluded from the model: *atemp* (near-perfect correlation with *temp*, $r = 0.99$), *mnth* (aliased with *season*), and *workingday* (linearly derived from *weekday* and *holiday*). Including aliased variables causes perfect multicollinearity, making VIF computation impossible due to a singular design matrix.

All remaining features returned VIF values below 3.6, indicating no problematic multicollinearity. The two highest values belong to *season* (3.54) and *temp* (3.41), which reflects a natural association between temperature and season and does not warrant further action.

4 Methodology

4.1 Ordinary Least Squares (OLS)

Ordinary Least Squares regression was selected as the primary model for its interpretability, providing coefficient estimates that directly quantify the effect of each predictor on daily rental counts. The model was trained using 5-fold cross-validation to obtain a reliable estimate of out-of-sample performance.

4.2 Regularisation

Ridge Regression Ridge regression extends OLS by adding an L2 penalty on the magnitude of coefficients, shrinking them towards zero to reduce variance. It retains all predictors in the model and is particularly useful when predictors are correlated. The penalty strength λ was selected via 5-fold cross-validation.

Lasso Regression Lasso regression applies an L1 penalty, which can shrink some coefficients to exactly zero, effectively performing feature selection. Both regularised models use the same cross-validation procedure for λ selection, and serve as a comparison against OLS to evaluate whether penalisation offers any predictive gain under the current feature set.

4.3 Evaluation Metrics

All three models were evaluated using 5-fold cross-validated RMSE as the primary metric, and R^2 as a supplementary measure of explained variance. The final models were trained on 8 predictors.

5 Assumption Diagnostics

5.1 Residuals vs. Fitted

Figure 8 plots residuals against fitted values to assess linearity and homoscedasticity. Residuals are roughly scattered around zero with no systematic U-shape or pattern, suggesting that the linearity assumption is largely satisfied. However, the spread of residuals appears slightly wider at higher fitted values, indicating mild heteroscedasticity.

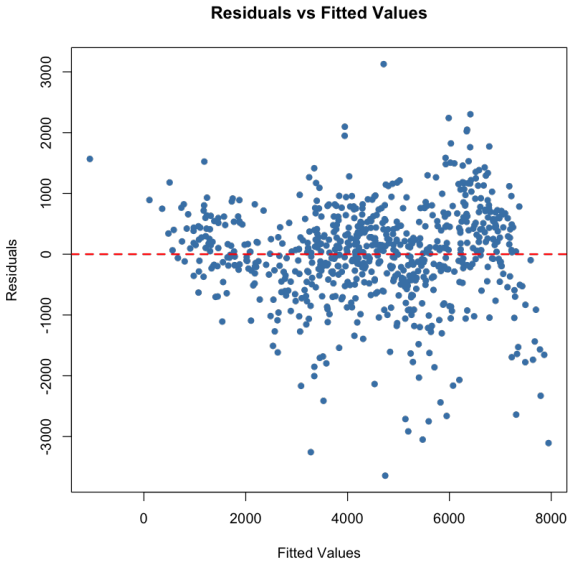

Figure 8: Residuals vs. Fitted Values

5.2 Normal Q–Q Plot

Figure 9 compares the distribution of residuals against a theoretical normal distribution. The middle section follows the reference line closely, indicating that residuals in the central range are approximately normally distributed. Both tails deviate from the line, suggesting slight heavy tails where extreme residuals are more frequent than a pure normal distribution would predict. One point in the upper right deviates notably and may represent an outlier. This pattern of mild leptokurtosis is common in real-world data and does not necessitate transformation.

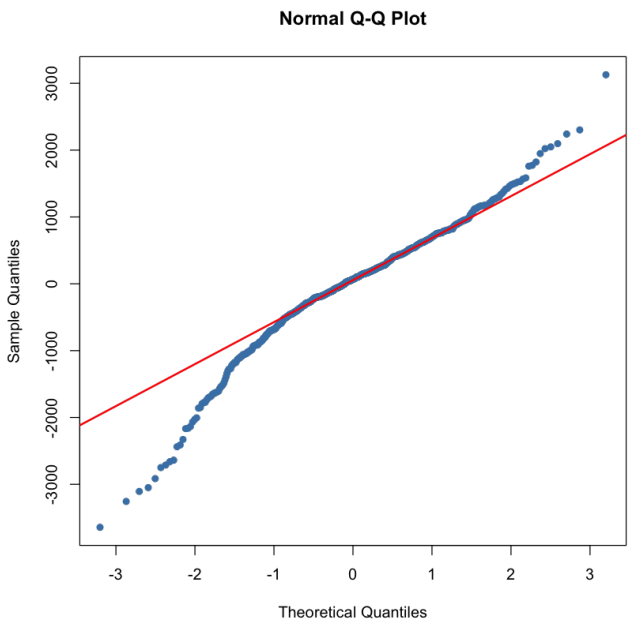

Figure 9: Normal Q–Q Plot of Residuals

5.3 Scale-Location

Figure 10 plots the square root of standardised residuals against fitted values to assess homoscedasticity. The red trend line shows a slight upward slope, indicating that residual variance increases mildly at higher fitted values. This is consistent with the pattern observed in the Residuals vs. Fitted plot and confirms mild heteroscedasticity, though the degree is not severe enough to require transformation.

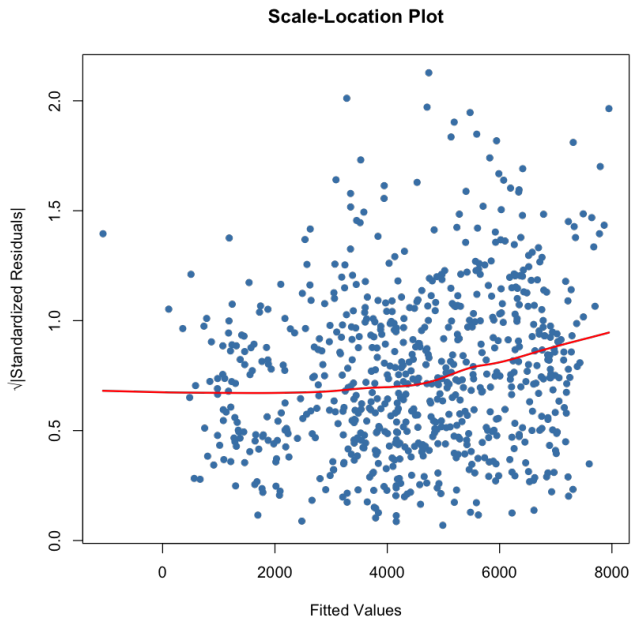

Figure 10: Scale-Location Plot

5.4 Cook's Distance

Figure 11 shows Cook's Distance for each observation, which measures how much the model coefficients would change if that observation were removed. Points exceeding the threshold of $4/n$ (red dashed line) are considered influential. A total of 39 observations (5.3%) exceed this threshold, distributed across the full observation period. This proportion is within a normal range and does not suggest that the model is being driven by a small number of anomalous points.

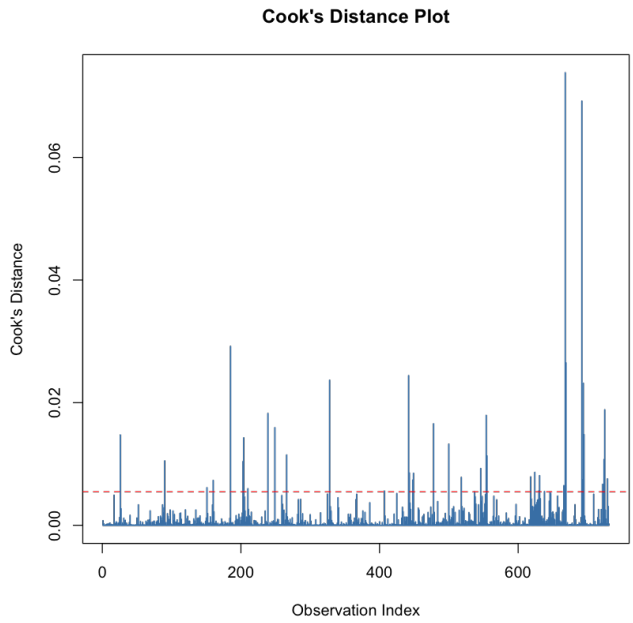

Figure 11: Cook's Distance

5.5 Summary

Overall, the linear model assumptions are largely satisfied. Linearity holds with no systematic pattern in the residuals. Mild heteroscedasticity is present at higher fitted values but is not severe enough to require transformation. Residuals show slight heavy tails but are approximately normal in the central range. Influential points account for 5.3% of observations, within an acceptable range.

6 Results

6.1 OLS Coefficient Estimates

The OLS model was trained on all 731 observations, achieving an in-sample R^2 of 0.827 and an adjusted R^2 of 0.824, indicating that approximately 82% of the variance in daily rentals is explained by the predictors.
Among the predictors, *yr* has the largest effect ($\hat{\beta} = 2017$, $t = 33.0$, $p < 0.001$), reflecting a strong year-over-year growth trend from 2011 to 2012. *temp* is the strongest continuous predictor ($\hat{\beta} = 5091$, $t = 16.7$, $p < 0.001$), confirming the positive relationship observed in the EDA. *windspeed* and *hum* both have significant negative effects, while *weathersit* deterioration is associated with substantially lower rentals. Among weekday indicators, *weekday1* is the only non-significant coefficient ($p = 0.07$), suggesting that Monday demand does not differ meaningfully from the Sunday baseline.

6.2 Model Comparison

Table 1 compares the three models using 5-fold cross-validated RMSE as the primary metric. OLS achieves the lowest CV RMSE of 821, outperforming Lasso (831) and Ridge

(835). The marginal difference suggests that regularisation offers limited improvement when the feature set is already well-specified.

Table 1: Model comparison. CV RMSE is out-of-sample (5-fold cross-validation); R^2 is in-sample.

Model	**CV RMSE**	R^2
OLS	821.99	0.827
Lasso	831.18	0.827
Ridge	834.62	0.824

7 Discussion

7.1 Key Findings

The OLS model explains approximately 82% of the variance in daily bike rentals, with all predictors significant except *weekday1*. Among all predictors, *yr* emerges as the dominant predictor, capturing a substantial year-over-year growth in demand from 2011 to 2012. However, this coefficient reflects a service growth trend rather than a stable seasonal or behavioural pattern, and may not generalise beyond the observation period. *temp* is the strongest continuous predictor, reinforcing the central role of weather conditions in driving rental demand. The comparison across models shows that OLS outperforms both Ridge and Lasso on CV RMSE (821 vs 831 vs 835), suggesting that regularisation offers limited benefit when the feature set is already well-specified through careful preprocessing.

7.2 Limitations

Three limitations are worth noting. First, the model covers only two years of data, and the large *yr* coefficient may reflect a growth phase specific to that period rather than a persistent trend, limiting the model's predictive validity for future years. Second, prediction errors are likely larger on days with extreme weather conditions, as such events are rare in the dataset and poorly represented during training. Third, the model does not include interaction terms, such as between *temp* and *season*, which may capture more nuanced demand dynamics and improve predictive performance.

References

[1] Fanaee-T, H. (2013). *Bike Sharing* [Dataset]. UCI Machine Learning Repository. `https://doi.org/10.24432/C5W894`